

produces `b/a/foo`, whereas the command:

```
rsync -a a/ b
```

produces `b/foo`. The presence or absence of a trailing slash on the destination argument (`b`, in this case) has no effect.

10.3 Using the `--delete` flag

If a file was originally in both `source/` and `destination/` (from an earlier `rsync`, for example), and you delete it from `source/`, you probably want it to be deleted from `destination/` on the next `rsync`. However, the default behavior is to leave the copy at `destination/` in place. Assuming you want `rsync` to delete any file from `destination/` that is not in `source/`, you'll need to use the `--delete` flag:

```
rsync -a --delete source/ destination/
```

10.4 Be lazy: use cron

One of the toughest obstacles to a good backup strategy is human nature; if there's any work involved, there's a good chance backups won't happen. Fortunately, there's a way to harness human laziness: make cron do the work.

To run the `rsync-with-backup` command from the previous section every morning at 4:20 AM, for example, edit the root cron table: (as root)

```
crontab -e
```

Then add the following line

```
20 4 * * * rsync -a --delete source/ destination/
```

Finally, save the file and exit. The backup will happen every morning at precisely 4:20 AM, and root will receive the output by email. Don't copy that example verbatim, though; you should use full path names (such as `/usr/bin/rsync` and `/home/source/`) to remove any ambiguity.

10.5 Incremental backups with `rsync`

Since making a full copy of a large filesystem can be a time-consuming and expensive process, it is common to make full backups only once a week or once a month, and store only changes on the other days. These are called "incremental" backups, and are supported by the venerable old `dump` and `tar` utilities, along with many others.

However, you don't have to use tape as your backup medium; it is both possible and vastly more efficient to perform incremental backups with `rsync`.

The most common way to do this is by using the `rsync -b --backup-dir=` combination. We've seen examples of that usage here, but we won't discuss it further, because there is a better way. If you're not familiar with